

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12387401
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Rui; Yuanle

Image processing method and electronic device

Abstract

An image processing method and an electronic device are provided. The image processing method includes: receiving a first input performed by a user; identifying a first text region on a target image in response to the first input; training a text image of the first text region on the target image to obtain a first font style model of the first text region; receiving a second input performed by the user; in response to the second input, obtaining a first input text, and training the first input text according to the first font style model to obtain a second text matching a font style of the first text region; and replacing text of the first text region with the second text on the target image.

Inventors:	Rui; Yuanle (Dongguan, CN)
Applicant:	VIVO MOBILE COMMUNICATION CO., LTD. (Dongguan, CN)
Family ID:	71432931
Assignee:	VIVO MOBILE COMMUNICATION CO., LTD. (Dongguan, CN)
Appl. No.:	17/952416
Filed:	September 26, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230018905 A1	Jan. 19, 2023

Foreign Application Priority Data

CN	202010229382.0	Mar. 27, 2020
----	----------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/CN2021/081983 20210322 PENDING child-doc US 17952416

Publication Classification

Int. Cl.: G06T11/60 (20060101); G06V30/19 (20220101); G06V30/414 (20220101)

U.S. Cl.:

CPC G06T11/60 (20130101); G06V30/1908 (20220101); G06V30/19147 (20220101); G06V30/414 (20220101);

Field of Classification Search

CPC: G06T (11/60); G06V (30/1908); G06V (30/19147); G06V (30/414)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0033874	12/2011	Perronnin et al.	N/A	N/A
2019/0243875	12/2018	Jain et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104715497	12/2014	CN	N/A
108241594	12/2017	CN	N/A
108829326	12/2017	CN	N/A
109189985	12/2018	CN	N/A
109285111	12/2018	CN	N/A
110008884	12/2018	CN	N/A
110458918	12/2018	CN	N/A
110738026	12/2019	CN	N/A
111402367	12/2019	CN	N/A

OTHER PUBLICATIONS

Wu, Editing Text in the Wild, arXiv:1908.03047v1 [cs.CV] Aug. 8, 2019 (Year: 2019). cited by examiner

Microsoft Community, How to change a selected font from whole document, <https://answers.microsoft.com/en-us/msoffice/forum/all/how-to-change-a-selected-font-from-whole-document/b3b47e54-c9e5-4e5c-be72-961ba3eda512>, 2018 (Year: 2018). cited by examiner

Extended European Search Report issued in related European Application No. 21776192.3, mailed Sep. 12, 2023, 6 pages. cited by applicant

Wu Liang et al, “Editing Text in the Wild”, Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, Oct. 15, 2019, 9 pages. cited by applicant

Prasun Roy et al: “STEFANN: Scene Text Editor using Font Adaptive Neural Network”, arxiv. org, Cornell University Library, 201 Olin Library Cornell University Ithaca, NY 14853, Mar. 4, 2019, 13 pages. cited by applicant

Office Action issued in related Korean Application No. 10-2022-7035671, mailed May 21, 2024, 5 pages. cited by applicant

Qiangpeng Yang, et al., “SwapText: Image Based Texts Transfer in Scenes”, CVPR 2020,

arXiv:2003.08152v1 [cs.CV] Mar. 18, 2020, 10 pages. cited by applicant
First Office Action issued in related Chinese Application No. 202010229382.0, mailed Mar. 11, 2023, 7 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/081983, mailed Jul. 1, 2021, 4 pages. cited by applicant
Examination report issued in related Indian Application No. 202217049908, mailed May 14, 2025, 8 pages. cited by applicant
Aksan et al., “DeepWriting: Making Digital Ink Editable via Deep Generative Modeling”, 14 pages. cited by applicant
Kuo et al., “HuayuNavi: A Mobile Chinese Learning Application Based on Intelligent Character Recognition” Dec. 11, 2011, 9 pages. cited by applicant
Office Action issued in related European Application No. 21776192.3, mailed Apr. 9, 2025, 4 pages. cited by applicant

Primary Examiner: Le; Vu

Assistant Examiner: Zhao; Lei

Attorney, Agent or Firm: IPX PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2021/081983, filed on Mar. 22, 2021, which claims priority to Chinese Patent Application No. 202010229382.0, filed on Mar. 27, 2020. The entire contents of each of the above-identified applications are expressly incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to the field of communication technologies, and in particular, to an image processing method and an electronic device.

BACKGROUND

(2) Currently, in order to perform picture text replacement editing by an electronic device, it is necessary to remove original text by performing a smearing operation, then add a text input box on the picture, set the color and font of the added text, and then paste the inputted text into the original picture.

(3) Although the current text editing method can realize text replacement, on the one hand, the user needs to manually smear and adjust the original text region on the picture, and in this process, it is difficult to control a smearing range in a refined manner, the complexity of the operation is high, and it is difficult to smear accurately; and on the other hand, only the default set font and color can be selected for the replacement text, and consequently, display effect inconsistency between the replacement text and the original text is prone to occur, making the text editing relatively rigid and causing a poor text editing effect.

SUMMARY

(4) The present disclosure provides an image processing method and an electronic device.

(5) According to a first aspect, an embodiment of the present disclosure provides an image processing method, performed by an electronic device, the method including:

(6) receiving a first input performed by a user;

(7) identifying a first text region on a target image in response to the first input;

(8) training a text image of a first text of the first text region on the target image to obtain a first

font style model of the first text;

(9) receiving a second input performed by the user;

(10) in response to the second input, obtaining a first input text, and training the first input text according to the first font style model to obtain a second text matching a font style of the first text;

(11) and replacing the first text with the second text on the target image.

(12) According to a second aspect, an embodiment of the present disclosure provides an electronic device, including:

(13) a first receiving module, configured to receive a first input performed by a user;

(14) an identification module, configured to identify a first text region on a target image in response to the first input;

(15) a first processing module, configured to train a text image of a first text of the first text region on the target image to obtain a first font style model of the first text;

(16) a second receiving module, configured to receive a second input performed by the user;

(17) a second processing module, configured to, in response to the second input, obtain a first input text, and train the first input text according to the first font style model to obtain a second text matching a font style of the first text; and

(18) a first replacement module, configured to replace the first text with the second text on the target image.

(19) According to a third aspect, an embodiment of the present disclosure provides an electronic device, including a processor, a memory, and a computer program stored on the memory and executable by the processor, the computer program implementing steps of the image processing method when executed by the processor.

(20) According to a fourth aspect, an embodiment of the present disclosure provides a computer-readable storage medium, where the computer-readable storage medium stores a computer program, the computer program implementing steps of the image processing method when executed by the processor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following describes the accompanying drawings for describing the embodiments of the present disclosure. Apparently, the accompanying drawings described below show merely some embodiments of the present disclosure, and a person of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.

(2) FIG. 1 shows a schematic flowchart of an image processing method according to an embodiment of the present disclosure;

(3) FIG. 2 shows a schematic structural diagram of an electronic device according to an embodiment of the present disclosure; and

(4) FIG. 3 shows a schematic structural diagram of hardware of an electronic device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(5) Referring to FIG. 1, FIG. 1 shows a schematic flowchart of an image processing method according to an embodiment of the present disclosure. An embodiment of the present disclosure provides an image processing method, applied to an electronic device having a screen display function, and the image processing method in this embodiment of the present disclosure may include the following steps.

(6) Step **101**: An electronic device receives a first input performed by a user.

(7) In this embodiment of the present disclosure, in a case that the electronic device displays a target image, the user may perform a first input operation to cause the electronic device to receive

the first input, thereby triggering the electronic device to perform an identification operation on a first text of the target image.

(8) The first input may include at least one of a voice input, a body action input, a touch input acting on the electronic device, or a motion input acting on the electronic device. The body action input may include, but is not limited to, at least one of a gesture action input, a head action input, a face action input, or the like, the touch input acting on the electronic device may include, but is not limited to, a touch input acting on a screen or a housing, and the motion input acting on the electronic device may include, but is not limited to, at least one of a swing action input and a flip action input acting on the electronic device, a curved input/bend input acting on a flexible screen, or the like.

(9) Step **102**: The electronic device identifies a first text region on a target image in response to the first input.

(10) In this embodiment of the present disclosure, in response to the first input received in step **101**, the electronic device detects the target image, and identifies the first text region displayed on the target image. In some embodiments, the electronic device can detect the target image by using the Optical Character Recognition (OCR) technology and Artificial Intelligence (AI) technology to identify the first text region on the target image and obtain the first text and a coordinate position of the first text on the target image.

(11) In some embodiments, in order to facilitate the user to intuitively determine an identification result, the electronic device can detect the content of the first text of the first text region on the target image and the coordinate position of the first text on the target image, and then mark and display the first text region on the target image by a predetermined marker. Exemplarily, the first text region on the target image can be marked and displayed by an image mask, for example, the first text region can be displayed by the image mask to a predetermined color (such as red) for the user to view and confirm. When the electronic device receives a movement input marked by the predetermined marker by the user, the electronic device performs text identification on an end position of the movement input in response to the movement input, to identify and determine the first text, thereby facilitating the user to correct a misidentified position in time, where the movement input may include at least one of a drag input or a press-drag input.

(12) Step **103**: The electronic device trains a text image of a first text of the first text region on the target image to obtain a first font style model of the first text.

(13) In this embodiment of the present disclosure, the electronic device inputs the text image of the first text identified in step **102** into a training model for AI training, thereby obtaining the first font style model corresponding to the first text, so that in a subsequent step, the font style can be kept unchanged in a process of performing text editing on the target image based on the first font style model, which enables the edited text to be seamlessly integrated with the target image.

(14) Step **104**: The electronic device receives a second input performed by the user.

(15) In this embodiment of the present disclosure, the user can trigger an editing modification to the first text by performing a second input operation. The second input may include at least one of a voice input, a body action input, a touch input acting on the electronic device, or a motion input acting on the electronic device.

(16) In some embodiments, in order to facilitate intuitive and quick operation by the user, the second input may be a preset touch input performed on the first text of the target image, and for example, the preset touch input may be at least one of a click input or a press input.

(17) Step **105**: In response to the second input, the electronic device obtains a first input text and trains the first input text according to the first font style model to obtain a second text matching a font style of the first text.

(18) In this embodiment of the present disclosure, in response to the second input received in step **104**, the electronic device obtains the first input text inputted by the user, and then inputs the first input text into the first font style model of the first text for training, thereby obtaining text matching

the font style of the first text, that is, the second text, so that a font style of the text inputted by the user and a font style of the original text (the first text) on the target image can remain unchanged, which enables the edited text (that is, the second text) to be seamlessly integrated with the target image.

(19) Step **106**. The electronic device replaces the first text with the second text on the target image.

(20) In this embodiment of the present disclosure, the electronic device replaces the first text on the target image with the second text obtained by training in step **105**, to conveniently perform the text replacement on the target image and enable the font style of the text inputted by the user and the font style of the original text on the target image to remain unchanged, so that the text inputted by the user can be seamlessly integrated with the target image.

(21) In some embodiments, after replacing the first text with the second text, the electronic device can integrate the second text with the target image for storage, thereby obtaining an image desired by the user.

(22) In the embodiments of the present disclosure, the electronic device receives a first input performed by a user, identifies a first text region on a target image in response to the first input, trains a text image of a first text of the first text region on the target image to obtain a first font style model, receives a second input, then trains, in response to the second input, an obtained first input text according to the first font style model to obtain a second text matching a font style of the first text, and replaces the first text with the second text on the target image. Therefore, text replacement performed on the target image can be conveniently performed, a font style of text inputted by the user and a font style of original text on the target image are kept unchanged, and the text inputted by the user is seamlessly integrated with the target image, thereby resolving the problem of complicated and rigid text replacement editing performed on the image.

(23) In some embodiments, in some embodiments of the present disclosure, step **103**, in which the electronic device trains a text image of a first text of the first text region on the target image to obtain a first font style model of the first text, may include the following steps: identifying, by the electronic device, a language category to which the first text belongs; and training, by the electronic device, the text image of the first text on the target image in a corpus corresponding to the language category to obtain the first font style model. In this way, the electronic device inputs the text image of the first text into a training model of the corpus corresponding to the language category to which the first text belongs for training, so that the font style model corresponding to the first text can be obtained based on quick and accurate training performed on corpus data of the corresponding corpus.

(24) In some embodiments, in some embodiments of the present disclosure, after step **106** in which the electronic device replaces the first text with the second text on the target image, the image processing method may further include: performing, by the electronic device, perspective transformation on the second text on the target image according to a mapping relationship between a layer on which the text image of the first text is located and a background layer in the target image. In this way, a placement setting position of the replaced second text on the target image can be consistent with a placement setting position of the original first text, which ensures that display effects of the text before and after editing are consistent.

(25) In some embodiments, in some embodiments of the present disclosure, after replacing the first text with the second text on the target image, the electronic device can further repair a text background region to restore an original background style effect. For example, after step **106** in which the electronic device replaces the first text with the second text on the target image, the image processing method may further include: generating, by the electronic device, a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text. In this way, new text (that is, the second text) can be seamlessly integrated with a background of the target image.

(26) For example, in some embodiments of the present disclosure, the generating, by the electronic

device, a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text may include one of the following.

(27) Method 1: The electronic device fills the region corresponding to the second text with pixels on the target image according to pixel information of a region corresponding to the text image of the first text, to generate the background image. In the method 1, the background image of the region corresponding to the second text is generated by using the pixel information of the region corresponding to the text image of the first text as a background filling basis, so that the background style of the region corresponding to the second text can be made consistent with the original background style of the first text.

(28) Method 2: The electronic device generates the background image in the region corresponding to the second text by using a generative adversarial networks model according to the pixel information of the region corresponding to the text image of the first text. In the method 2, a generative model and a discriminative model of the Generative Adversarial Networks (GANs) model are used to learn the pixel information of the region corresponding to the text image of the first text by gaming with each other, so as to generate the background image of the region corresponding to the second text, so that the background style of the region corresponding to the second text can be made more consistent with the original background style of the first text region.

(29) In this embodiment of the present disclosure, after step **106** in which the electronic device replaces the first text with the second text on the target image, the electronic device can also generate the background image through the pixel information around the region where the second text is located.

(30) In some embodiments, in some embodiments of the present disclosure, a translation function can be further provided in a process of modifying and editing the text. The electronic device can translate the text inputted by the user into a preset language type and then input it into the first font style model for training. That is, in the embodiments of the present disclosure, the first input text may be text inputted by the second input, or may be translation text of the text inputted by the second input translated into the preset language type. In this way, the diversified editing needs of the user can be met.

(31) In some embodiments, in some embodiments of the present disclosure, in order to further realize the diversified editing of text in the target image, after step **102** in which the electronic device identifies a first text region on a target image in response to the first input, the image processing method may further include the following steps: displaying, by the electronic device, preset stored font styles to be selected; receiving, by the electronic device, a third input of a target font style in the font styles to be selected by the user; training, by the electronic device in response to the third input, the first text according to a second font style model corresponding to the target font style to obtain a third text matching the target font style; and replacing, by the electronic device, the first text with the third text on the target image. This makes it easy for the user to apply their preferred font styles to modify and edit the text of the target image. The font style to be selected may be a font style pre-stored or pre-collected by the electronic device, or may be a font style used in a predetermined time period on the electronic device. In addition, the electronic device stores a font style model corresponding to the font style to be selected, for example, the predetermined time period may be within a week before the current system time, and the font style to be selected may be a font style used within a week (that is, used recently). In addition, in order to avoid occupying excessive cache resources of the electronic device, in a case that the font style to be selected is a font style used in a predetermined time period on the electronic device, the number of the font styles to be selected may be predetermined, and the predetermined number may be set by default by the electronic device, or may be set by the user, for example, the predetermined number may be three.

(32) For example, in some embodiments of the present disclosure, in order to facilitate the user to

quickly perform an editing operation on the first text, after step **103** in which the electronic device trains a text image of a first text on the target image to obtain a first font style model of the first text, the image processing method may further include: displaying the first text in a text box on the target image. In this way, the user can implement quick text editing based on the first text displayed in a form of the text box. In some embodiments, the electronic device may mark the first text region with a quadrilateral on the target image and then perform perspective transformation on the quadrilateral marker to display the first text in the text box.

(33) Exemplarily, step **104**, in which the electronic device receives a second input performed by the user, may include: receiving the second input performed on the text box, where the second input may be a preset touch input, for example, may be at least one of a click input performed on the text box or a press input performed on the text box. Further, in order to facilitate the user to quickly edit and input, in step **105**, before the first input text is obtained and the first input text is trained according to the first font style model to obtain the second text matching the font style of the first text, the electronic device displays a text input interface in response to the second input, and the electronic device extracts text content of the first text and displays the text content of the first text on the text input interface, so that the user can quickly perform editing and modification again on the text content of the first text through the text input interface. The text input interface may be an input keyboard.

(34) Exemplarily, after the electronic device displays the first text in the text box on the target image, the electronic device may receive a preset input performed on the text box by the user, and adjust at least one of the display size or display position of the first text on the target image in response to the preset input, for example, the user can zoom in, rotate, and move the first text through the preset input performed the text box.

(35) Exemplarily, in order to improve the human-machine experience of the user and facilitate the user to operate simply, in this embodiment of the present disclosure, a preset control may be configured, and before the electronic device operates the corresponding adjustment input (such as the first input, the second input, or the third input), a touch input performed on the preset control by the user may be received, so as to trigger the adjustment input to enable a text replacement function, a translation function, and a function of adjusting the display size and the display position in the editing operation performed on the first text, thereby avoiding mistaken operations of the user.

(36) In the image processing method according to the embodiments of the present disclosure, a first input performed by a user is received; a first text region on a target image is identified in response to the first input; a text image of a first text of the first text region on the target image is trained to obtain a first font style model; a second input is received; then in response to the second input, an obtained first input text is trained according to the first font style model, to obtain a second text matching a font style of the first text; and the first text is replaced with the second text on the target image. Therefore, text replacement performed on the target image can be conveniently implemented, a font style of text inputted by the user and a font style of original text on the target image are kept unchanged, and the text inputted by the user is seamlessly integrated with the target image, thereby resolving the problem of complicated and rigid text replacement editing performed on the image.

(37) Based on the above method, an embodiment of the present disclosure provides an electronic device configured to implement the above method.

(38) Referring to FIG. 2, FIG. 2 shows a schematic structural diagram of an electronic device according to an embodiment of the present disclosure. An embodiment of the present disclosure provides an electronic device **200**, and the electronic device **200** may include: a first receiving module **210**, an identification module **220**, a first processing module **230**, a second receiving module **240**, a second processing module **250**, and a first replacement module **260**.

(39) The first receiving module **210** is configured to receive the first input performed by a user;

- (40) the identification module **220** is configured to identify a first text region on a target image in response to the first input;
- (41) the first processing module **230** is configured to train a text image of a first text of the first text region on the target image to obtain a first font style model of the first text;
- (42) the second receiving module **240** is configured to receive a second input performed by the user;
- (43) the second processing module **250** is configured to, in response to the second input, obtain a first input text, and train the first input text according to the first font style model to obtain a second text matching a font style of the first text; and
- (44) the first replacement module **260** is configured to replace the first text with the second text on the target image.
- (45) In some embodiments, in some embodiments of the present disclosure, the first processing module **230** may include: an identification unit and a first processing unit.
- (46) The identification unit is configured to identify a language category to which the first text belongs; and
- (47) the first processing unit is configured to train the text image of the first text on the target image in a corpus corresponding to the language category to obtain the first font style model.
- (48) In some embodiments, in some embodiments of the present disclosure, the electronic device **200** may further include: a third processing module.
- (49) The third processing module is configured to perform perspective transformation on the second text on the target image according to a mapping relationship between a layer on which the text image of the first text is located and a background layer in the target image.
- (50) In some embodiments, in some embodiments of the present disclosure, the electronic device **200** may further include: a fourth processing module.
- (51) The fourth processing module is configured to generate a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text.
- (52) For example, in some embodiments of the present disclosure, the fourth processing module may include one of the following: a second processing unit or a third processing unit.
- (53) The second processing unit is configured to fill the region corresponding to the second text with pixels on the target image according to pixel information of a region corresponding to the text image of the first text, to generate the background image; or
- (54) the third processing unit is configured to generate the background image in the region corresponding to the second text by using a generative adversarial networks model according to the pixel information of the region corresponding to the text image of the first text.
- (55) For example, in some embodiments of the present disclosure, the first input text is the text inputted by the second input, or the translation text of the text inputted by the second input translated into the preset language type.
- (56) For example, in some embodiments of the present disclosure, the electronic device **200** may further include: a display module, a third receiving module, a fourth processing module, and a second replacement module.
- (57) The display module is configured to display preset stored font styles to be selected;
- (58) the third receiving module is configured to receive a third input of a target font style in the font styles to be selected by the user;
- (59) the fourth processing module is configured to, train, in response to the third input, the first text according to a second font style model corresponding to the target font style to obtain a third text matching the target font style; and
- (60) the second replacement module is configured to replace the first text with the third text on the target image.
- (61) The electronic device **200** according to this embodiment of the present disclosure can

implement various processes implemented by the electronic device in the method embodiment in FIG. 1, and the description is not repeated here in order to avoid repetition.

(62) By using the electronic device according to the embodiments of the present disclosure, under cooperation of the first receiving module, the identification module, and the first processing module, a first input performed by a user is received; a first text region on a target image is identified in response to the first input, a text image of a first text of the first text region on the target image is trained to obtain a first font style model; and then under cooperation of the second receiving module, the second processing module, and the first replacement module, a second input is received; in response to the second input, an obtained first input text is trained according to the first font style model, to obtain a second text matching a font style of the first text; and the first text is replaced with the second text on the target image. Therefore, text replacement performed on the target image can be conveniently implemented, a font style of text inputted by the user and a font style of original text on the target image are kept unchanged, and the text inputted by the user is seamlessly integrated with the target image, thereby resolving the problem of complicated and rigid text replacement editing performed on the image.

(63) FIG. 3 is a schematic structural diagram of hardware of an electronic device implementing various embodiments of the present disclosure.

(64) This electronic device **300** includes, but is not limited to, components such as a radio frequency unit **301**, a network module **302**, an audio output unit **303**, an input unit **304**, a sensor **305**, a display unit **306**, a user input unit **307**, an interface unit **308**, a memory **309**, a processor **310**, and a power supply **311**. A person skilled in the art may understand that the electronic device structure shown in FIG. 3 does not constitute a limitation to the electronic device. The electronic device may include more or fewer components than those shown in the figure, or may combine some components, or may have different component arrangements. In this embodiment of the present disclosure, the electronic device includes, but is not limited to, a mobile phone, a tablet computer, a laptop computer, a handheld computer, an in-vehicle terminal, a wearable device, a pedometer, and the like.

(65) The user input unit **307** is configured to receive a first input performed by a user; the processor **310** is configured to identify a first text region on a target image in response to the first input; and train a text image of a first text of the first text region on the target image to obtain a first font style model of the first text. The user input unit **307** is further configured to receive a second input performed by the user; the processor **310** is further configured to, in response to the second input, obtain the first input text, and train the first input text according to the first font style model to obtain a second text matching a font style of the first text; and the display unit **306** is configured to replace the first text with the second text on the target image. In this way, the text replacement performed on the target image can be conveniently implemented, the font style of the text inputted by the user and the font style of the original text on the target image are kept unchanged, and the text inputted by the user is seamlessly integrated with the target image, thereby resolving the problem of complicated and rigid text replacement editing performed on the image.

(66) It should be understood that in this embodiment of the present disclosure, the radio frequency unit **301** may be configured to send and receive signals in an information receiving and sending process or a call process. In some embodiments, the radio frequency unit **301** is configured to receive downlink data from a base station, and then transmit the downlink data to the processor **310** for processing; and additionally, send uplink data to the base station. Generally, the radio frequency unit **301** includes, but is not limited to, an antenna, at least one amplifier, a transceiver, a coupler, a low noise amplifier, a duplexer, or the like. In addition, the radio frequency unit **301** may further communicate with another device through a wireless communication system and a network.

(67) The electronic device provides wireless broadband Internet access to the user through the network module **302**, for example, helps the user to send and receive emails, browse web pages, and access stream media.

(68) The audio output unit **303** may convert audio data received by the radio frequency unit **301** or the network module **302** or audio data stored in the memory **309** into audio signals and output the audio signals as sounds. Moreover, the audio output unit **303** may further provide an audio output related to a specific function executed by the electronic device **300** (for example, a call signal reception sound, and a message reception sound). The audio output unit **303** includes a speaker, a buzzer, a receiver, and the like.

(69) The input unit **304** is configured to receive audio or video signals. The input unit **304** may include a Graphics Processing Unit (GPU) **3041** and a microphone **3042**. The graphics processing unit **3041** processes image data of static pictures or videos obtained by an image capture device (for example, a camera) in a video capture mode or an image capture mode. The processed image frames may be displayed on the display unit **306**. The image frames processed by the graphics processing unit **3041** may be stored in the memory **309** (or another storage medium) or sent via the radio frequency unit **301** or the network module **302**. The microphone **3042** may receive sounds and can process such sounds as audio data. The processed audio data may be converted, in a phone talk mode, into a format that may be sent to a mobile communication base station via the radio frequency unit **301** for output.

(70) The electronic device **300** further includes at least one sensor **305**, such as an optical sensor, a motion sensor, or other sensors. In some embodiments, the optical sensor includes an ambient light sensor and a proximity sensor. The ambient light sensor may adjust luminance of a display panel **3061** according to brightness of the ambient light. The proximity sensor may switch off the display panel **3061** and/or backlight when the electronic device **300** is moved to ears. As one type of motion sensor, an accelerometer sensor may detect magnitude of accelerations in various directions (generally, on three axes), may detect magnitude and a direction of gravity when static, and may be configured to recognize the attitude of the electronic device (for example, switching between landscape orientation and portrait orientation, a related game, and magnetometer attitude calibration), a function related to vibration recognition (such as a pedometer and a knock), and the like. The sensor **305** may further include a fingerprint sensor, a pressure sensor, an iris sensor, a molecular sensor, a gyroscope, a barometer, a hygrometer, a thermometer, an infrared sensor, and the like. Details are not described herein again.

(71) The display unit **306** is configured to display information inputted by the user or information provided to the user. The display unit **306** may include a display panel **3061**, and the display panel **3061** may be configured in the form of a Liquid Crystal Display (LCD), an Organic Light-Emitting Diode (OLED), or the like.

(72) The user input unit **307** may be configured to receive input digit or character information, and generate a keyboard signal input related to the user setting and function control of the electronic device. In some embodiments, the user input unit **307** may include a touch panel **3071** and another input device **3072**. The touch panel **3071**, also referred to as a touchscreen, may collect a touch operation performed by a user on or near the touch panel (such as an operation performed by a user on or near the touch panel **3071** by using any suitable object or attachment, such as a finger or a touch pen). The touch panel **3071** may include two parts: a touch detection apparatus and a touch controller. The touch detection apparatus detects a touch orientation of the user, detects a signal brought by the touch operation, and transmits the signal to the touch controller. The touch controller receives touch information from the touch detection apparatus, converts the touch information into contact coordinates, then transmits the contact coordinates to the processor **310**, and receives and executes a command transmitted by the processor **310**. In addition, the touch panel **3071** may be implemented by using various types, such as a resistive type, a capacitive type, an infrared type, and a surface acoustic wave type. In addition to the touch panel **3071**, the user input unit **307** may further include the another input device **3072**. In some embodiments, the another input device **3072** may include, but is not limited to, a physical keyboard, a functional key (such as a volume control key or a switch key), a track ball, a mouse, and a joystick, and details are

not described herein again.

(73) Further, the touch panel **3071** may cover the display panel **3061**. After detecting a touch operation on or near the touch panel **3071**, the touch panel **3071** transfers the touch operation to the processor **310**, so as to determine a type of the touch event. Then, the processor **310** provides corresponding visual output on the display panel **3061** according to the type of the touch event. Although, in FIG. 3, the touch panel **3071** and the display panel **3061** are used as two separate parts to implement input and output functions of the electronic device, in some embodiments, the touch panel **3071** and the display panel **3061** may be integrated to implement the input and output functions of the electronic device, and details are not described herein again.

(74) The interface unit **308** is an interface for connection between an external device and the electronic device **300**. For example, the external device may include a wired or wireless headphone port, an external power (or a battery charger) port, a wired or wireless data port, a memory card port, a port for connecting devices with an identification module, an audio Input/Output (I/O) port, a video I/O port, a headphone port, and the like. The interface unit **308** may be configured to receive input from the external device (for example, data information and power) and transmit the received input to one or more elements in the electronic device **300**, or may be configured to transmit data between the electronic device **300** and the external device.

(75) The memory **309** may be configured to store software programs and various pieces of data. The memory **309** may mainly include a program storage region and a data storage region. The program storage region may store an operating system, an application program required by at least one function (for example, a sound playback function and an image playback function), or the like. The data storage region may store data (for example, audio data and a phone book) created according to use of the mobile phone. In addition, the memory **309** may include a high-speed random access memory, and may further include a non-volatile memory, such as at least one magnetic disk storage device, a flash memory device, or other non-volatile solid state storage devices.

(76) The processor **310** is a control center of the electronic device, and connects various parts of the entire electronic device by using various interfaces and lines. By running or executing a software program and/or a module stored in the memory **309**, and invoking data stored in the memory **309**, the processor performs various functions of the electronic device and processes data, thereby performing overall monitoring on the electronic device. The processor **310** may include one or more processing units. In some embodiments, the processor **310** may integrate an application processor and a modem processor. The application processor mainly processes an operating system, a user interface, an application program, and the like. The modem processor mainly processes wireless communication. It may be understood that the modem processor may either not be integrated into the processor **310**.

(77) The electronic device **300** may further include the power supply **311** (such as a battery) for supplying power to the components. In some embodiments, the power supply **311** may be logically connected to the processor **310** by using a power management system, thereby implementing functions such as charging, discharging, and power consumption management by using the power management system.

(78) In addition, the electronic device **300** includes a number of functional modules that are not shown, and details are not described herein again.

(79) For example, an embodiment of the present disclosure further provides an electronic device, including a processor **310**, a memory **309**, and a computer program stored on the memory **309** and executable by the processor **310**. The computer program, when executed by the processor **310**, implements various processes of the embodiments of the image processing method, and the same technical effect can be achieved. In order to avoid repetition, details are not described herein again.

(80) An embodiment of the present disclosure further provides a computer-readable storage medium, storing a computer program. When executed by a processor, the computer program

implements various processes of the embodiments of the image processing method, and the same technical effect can be achieved. In order to avoid repetition, details are not described herein again. The computer-readable storage medium is, for example, a Read-Only Memory (ROM), a Random Access Memory (RAM), a magnetic disk, or an optical disc.

(81) It may be understood that these embodiments described in the present disclosure may be implemented by hardware, software, firmware, middleware, microcode, or a combination thereof. For hardware implementation, modules, units, sub-modules, sub-units, and the like may be implemented in one or more Application Specific Integrated Circuit (ASIC), Digital Signal Processor (DSP), DSP Device (DSPD), Programmable Logic Device (PLD), Field-Programmable Gate Array (FPGA), general purpose processors, controllers, microcontrollers, microprocessors, other electronic units for performing the functions described in this application, or combinations thereof.

(82) It should be noted that, in this specification, the term “include”, “comprise”, or any other variant is intended to cover a non-exclusive inclusion, so that a process, a method, an article, or an apparatus including a series of elements not only includes those elements, but also includes other elements that are not listed explicitly, or may further include elements inherent in the process, the method, the article, or the apparatus. Without more limitations, elements defined by the sentence “including one” does not exclude that there are still other same elements in the process, the method, the article, or the apparatus.

(83) According to the description of the above implementations, a person skilled in the art may clearly understand that the method according to the embodiment may be implemented by means of software and a necessary general hardware platform, and certainly, may also be implemented by hardware, but in many cases, the former manner is a better implementation. Based on such an understanding, the technical solutions of the present disclosure essentially, or the part contributing to the prior art, may be presented in the form of a software product. The computer software product is stored in a storage medium (for example, a ROM/RAM, a magnetic disk, or an optical disc), and contains several instructions to enable a terminal (which may be a mobile phone, a computer, a server, an air conditioner, a network device, or the like) to perform the methods according to the embodiments of the present disclosure.

(84) The embodiments of the present disclosure have been described above with reference to the accompanying drawings, but the present disclosure is not limited to the above specific embodiments. The above specific embodiments are merely illustrative but not restrictive. A person of ordinary skill in the art may make various variations under the enlightenment of the present disclosure without departing from the spirit of the present disclosure and the protection scope of the claims, and such variations shall all fall within the protection scope of the present disclosure.

Claims

1. An image processing method, performed by an electronic device, wherein the method comprises: receiving a first input performed by a user; identifying a first text region on a target image in response to the first input; training a first font style model of a first text of the first text region on the target image using a text image of the first text; receiving a second input performed by the user; in response to the second input, obtaining a first input text, and applying the first font style model to obtain a second text matching a font style of the first text based on the first input text; and replacing the first text with the second text on the target image, wherein after replacing the first text with the second text on the target image, the method further comprises: performing perspective transformation on the second text on the target image according to a mapping relationship between a layer on which the text image of the first text is located and a background layer in the target image.

2. The method according to claim 1, wherein training the first font style model of the first text

comprises: identifying a language category to which the first text belongs; and training the first font style model using a corpus corresponding to the language category.

3. The method according to claim 1, wherein after the replacing the first text with the second text on the target image, the method further comprises: generating a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text.

4. The method according to claim 3, wherein the generating a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text comprises one of the following: filling the region corresponding to the second text with pixels on the target image according to pixel information of a region corresponding to the text image of the first text, to generate the background image; or generating the background image in the region corresponding to the second text by using a generative adversarial networks model according to the pixel information of the region corresponding to the text image of the first text.

5. The method according to claim 1, wherein after the identifying a first text region on a target image in response to the first input, the method further comprises: displaying preset stored font styles to be selected; receiving a third input of a target font style in the font styles to be selected by the user; training, in response to the third input, a second font style model corresponding to the target font style to obtain a third text matching the target font style using the first text; and replacing the first text with the third text on the target image.

6. An electronic device, comprising: a memory storing computer-readable instructions; and a processor coupled to the memory and configured to execute the computer-readable instructions, wherein the computer-readable instructions, when executed by the processor, cause the processor to perform operations comprising: receiving a first input performed by a user; identifying a first text region on a target image in response to the first input; training a first font style model of a first text of the first text region on the target image using a text image of the first text; receiving a second input performed by the user; in response to the second input, obtaining a first input text, and applying the first font style model to obtain a second text matching a font style of the first text based on the first input text; and replacing the first text with the second text on the target image, wherein after replacing the first text with the second text on the target image, the operations further comprise: performing perspective transformation on the second text on the target image according to a mapping relationship between a layer on which the text image of the first text is located and a background layer in the target image.

7. The electronic device according to claim 6, wherein training the first font style model of the first text comprises: identifying a language category to which the first text belongs; and training the first font style model using a corpus corresponding to the language category.

8. The electronic device according to claim 6, wherein after the replacing the first text with the second text on the target image, the operations further comprise: generating a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text.

9. The electronic device according to claim 8, wherein the generating a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text comprises one of the following: filling the region corresponding to the second text with pixels on the target image according to pixel information of a region corresponding to the text image of the first text, to generate the background image; or generating the background image in the region corresponding to the second text by using a generative adversarial networks model according to the pixel information of the region corresponding to the text image of the first text.

10. The electronic device according to claim 6, wherein after the identifying a first text region on a target image in response to the first input, the operations further comprise: displaying preset stored

font styles to be selected; receiving a third input of a target font style in the font styles to be selected by the user; training, in response to the third input, a second font style model corresponding to the target font style to obtain a third text matching the target font style using the first text; and replacing the first text with the third text on the target image.

11. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform operations comprising: receiving a first input performed by a user; identifying a first text region on a target image in response to the first input; training a first font style model of a first text of the first text region on the target image using a text image of the first text; receiving a second input performed by the user; in response to the second input, obtaining a first input text, and applying the first font style model to obtain a second text matching a font style of the first text based on the first input text; and replacing the first text with the second text on the target image, wherein after replacing the first text with the second text on the target image, the operations further comprise: performing perspective transformation on the second text on the target image according to a mapping relationship between a layer on which the text image of the first text is located and a background layer in the target image.

12. The non-transitory computer-readable medium according to claim 11, wherein training the first font style model of the first text comprises: identifying a language category to which the first text belongs; and training the first font style model using a corpus corresponding to the language category.

13. The non-transitory computer-readable medium according to claim 11, wherein after the replacing the first text with the second text on the target image, the operations further comprise: generating a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text.

14. The non-transitory computer-readable medium according to claim 13, wherein the generating a background image in a region corresponding to the second text on the target image according to a background style corresponding to the text image of the first text comprises one of the following: filling the region corresponding to the second text with pixels on the target image according to pixel information of a region corresponding to the text image of the first text, to generate the background image; or generating the background image in the region corresponding to the second text by using a generative adversarial networks model according to the pixel information of the region corresponding to the text image of the first text.

15. The non-transitory computer-readable medium according to claim 11, wherein after the identifying a first text region on a target image in response to the first input, the operations further comprise: displaying preset stored font styles to be selected; receiving a third input of a target font style in the font styles to be selected by the user; training, in response to the third input, a second font style model corresponding to the target font style to obtain a third text matching the target font style using the first text; and replacing the first text with the third text on the target image.
